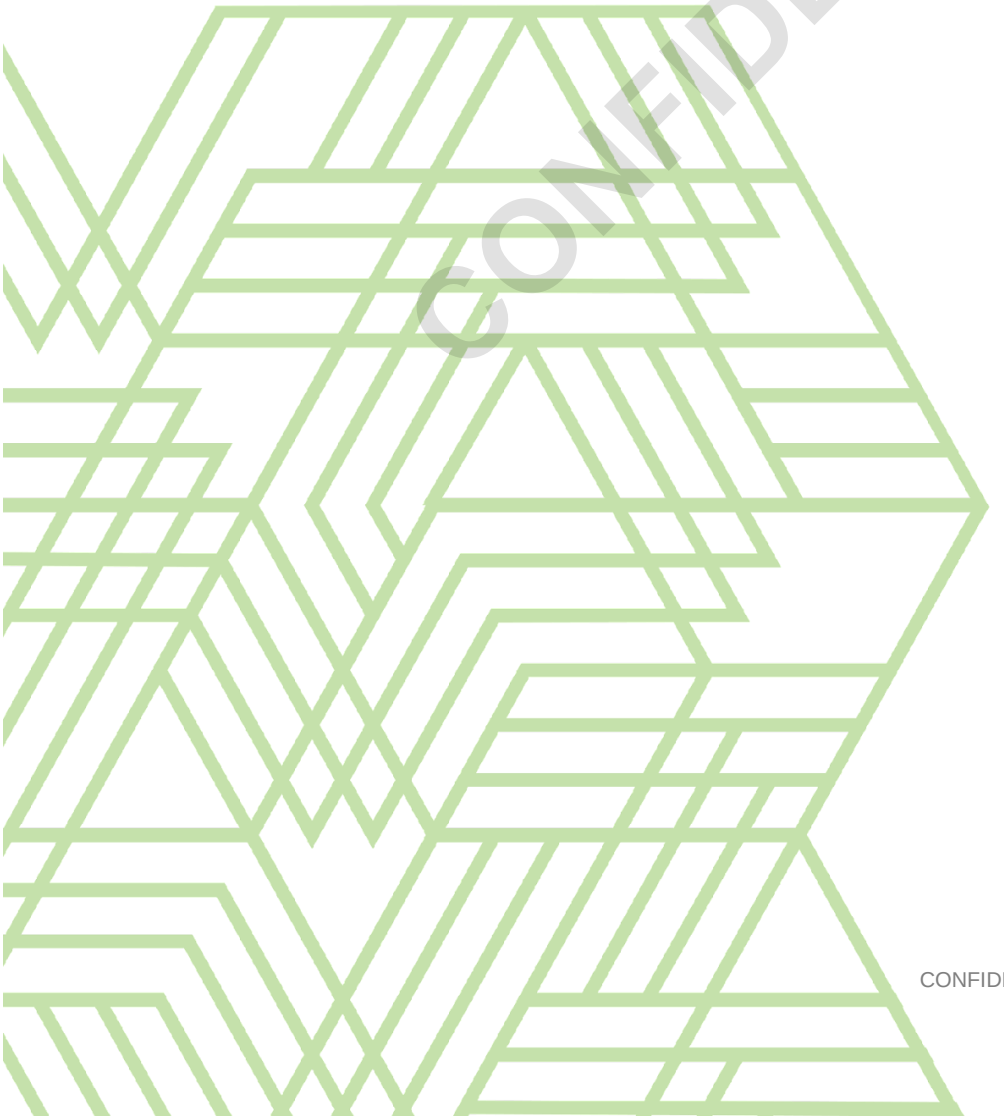
ATM3231 PV Remote Control

User Guide

SUMMARY: This document describes the features and usage of the ATM3231 based Photovoltaic (PV) Bluetooth LE Remote Control Reference Design. The Remote Control is powered by a PV cell and is available in two energy storage configurations: coin cell battery + supercapacitor, and battery free.

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Acronyms and Abbreviations

Acronyms	Definition
ADPCM	Adaptive Differential Pulse-Code Modulation
IMA	Interactive Multimedia Association
IR	Infrared
SoC	System on Chip
OTA	Over The Air
PMU	Power Management Unit
PV	Photovoltaic
RCU	Remote Control Unit

1. Overview

The ATM3231 PV Remote Control is a Bluetooth LE TV Remote Control reference design that demonstrates the low power consumption of the Atmosic Bluetooth LE ATM3231 and how addition of a PV cell to a remote control can reduce or eliminate the need for batteries in the device.

This document covers the features, operation, performance, and upgrade procedure for the ATM3231 PV Remote Control.

The ATM3231 PV Remote Control comes in two different configurations based on how harvested energy is stored in the device. The ATMRC-3231-HSC stores harvested energy in a hybrid supercapacitor that can also be recharged via a USB connector on the device. The ATMRC-3231-CC model has an integrated non-rechargeable CR2032 coin cell battery and a supercapacitor for harvested energy storage.

The ATM3231 SoC solution combines an ultra low power Bluetooth LE 5.0 solution with an integrated PMU that supports the direct connection of PV cells.

The direct connection of a PV cell to the ATM3231 PMU has multiple benefits, both reducing the overall cost of adding energy harvesting and improving the efficiency of harvested energy usage. In addition, the low power consumption of the ATM3231 means less energy needs to be harvested, so a smaller PV cell can be used in the design, further reducing both the cost and area required for the PV cell.

The ATM3231 PV Remote Control is designed to operate at indoor light levels with excess harvested energy stored in the supercapacitor or hybrid supercapacitor.

The CR2032 coin cell battery model only uses the battery when both harvested and stored energy is not available, such as during extended periods of operation in a dark environment. The ATM3231 PMU always uses harvested and stored energy whenever possible to extend the battery life of this Remote Control design.

The ATM3231 PV Remote Control is Android12 compatible and has been verified to work with the ADT-3 Android TV Developer Kit from Google and the onn. UHD Streaming Device.

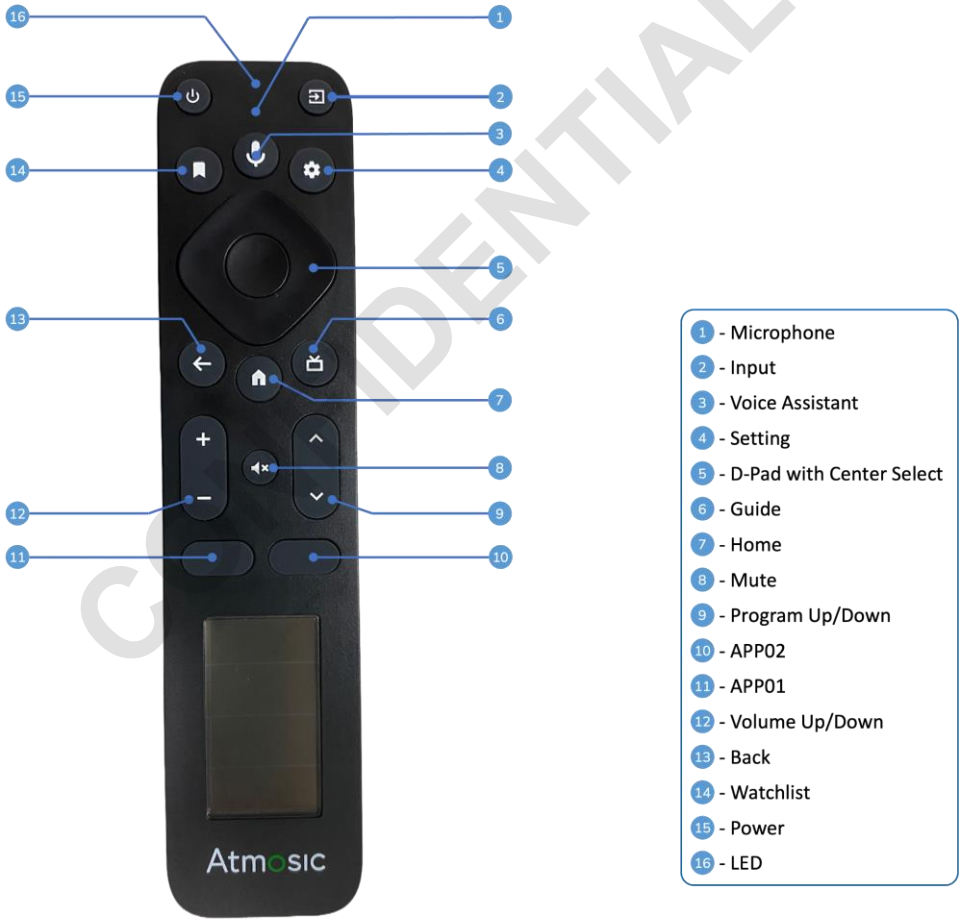
The Atmosic PV Remote Control is provided for demonstration and evaluation purposes only and is not for commercial sale.

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2. PV Remote Control

The ATM3231 PV Remote Control is shown in Figure 1. The PV cell on the front is an AM-1454 amorphous silicon PV cell from Panasonic. The Remote Control also supports the following features:

- Android 12 TV
- Google Voice over Bluetooth LE (version 0.4e/1.0)
- 8 KHz/16-bit IMA ADPCM voice encoder
- NEC IR Transmission Protocol
- 20 keys (5x4 matrix)



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Figure 1 - ATM3231 PV Remote Control, Front

Internally, there are two energy storage configurations. The coin cell configuration (ATMRC-3231-CC) hosts a CR2032 battery and a 0.5 F supercapacitor for storage of harvested energy. The battery free configuration (ATMRC-3231-HSC) integrates two 50 F hybrid supercapacitors (see References section for more information). The

internal hardware of the Remote Control is shown in Figure 2. The CR2032 battery or hybrid supercapacitors can be accessed by removing the back cover.

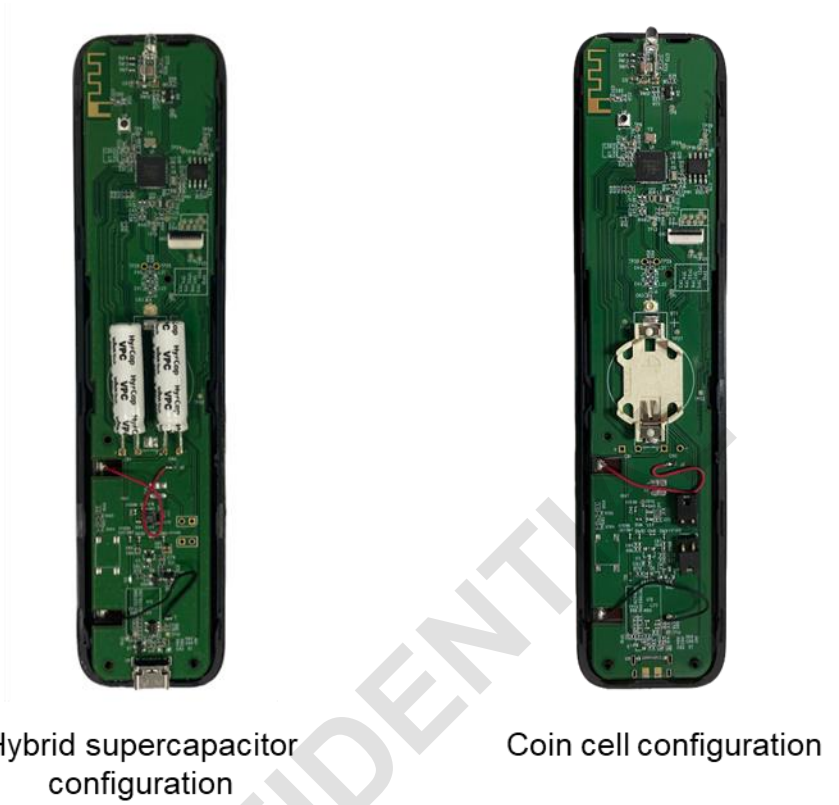


Figure 2 - ATM3231 PV Remote Control Power Options with Back Cover Removed

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The design of the two remote models are almost identical with one exception being the energy storage. Figure 3 below depicts the main components of the coin cell configuration while the main components of the hybrid supercapacitor based solution are shown in Figure 4. More information about the 0.5 F supercapacitor and 50 F hybrid supercapacitors can be found in the References section.

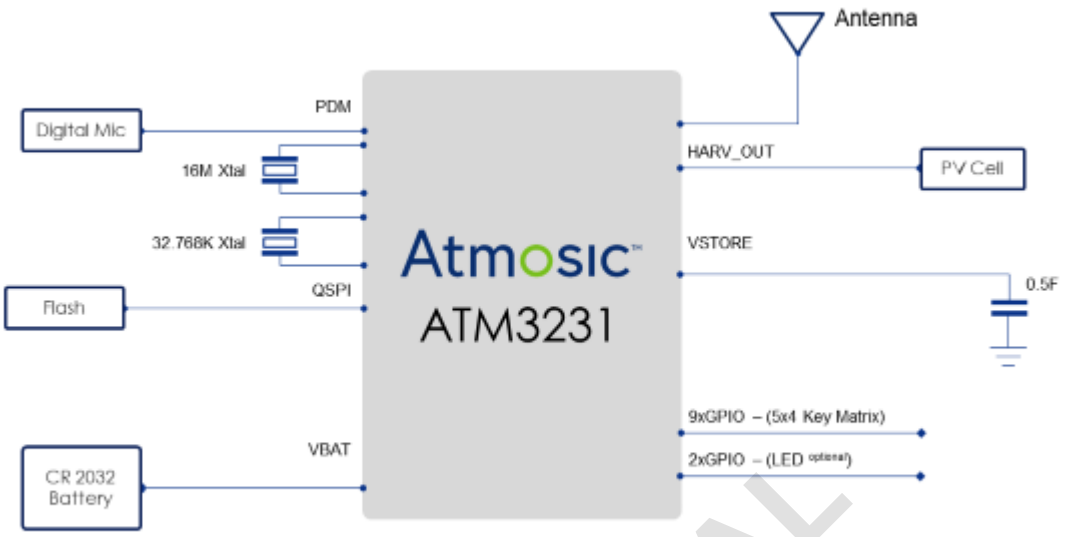


Figure 3 - Block Diagram for CR2032 Coin Cell Battery Configuration

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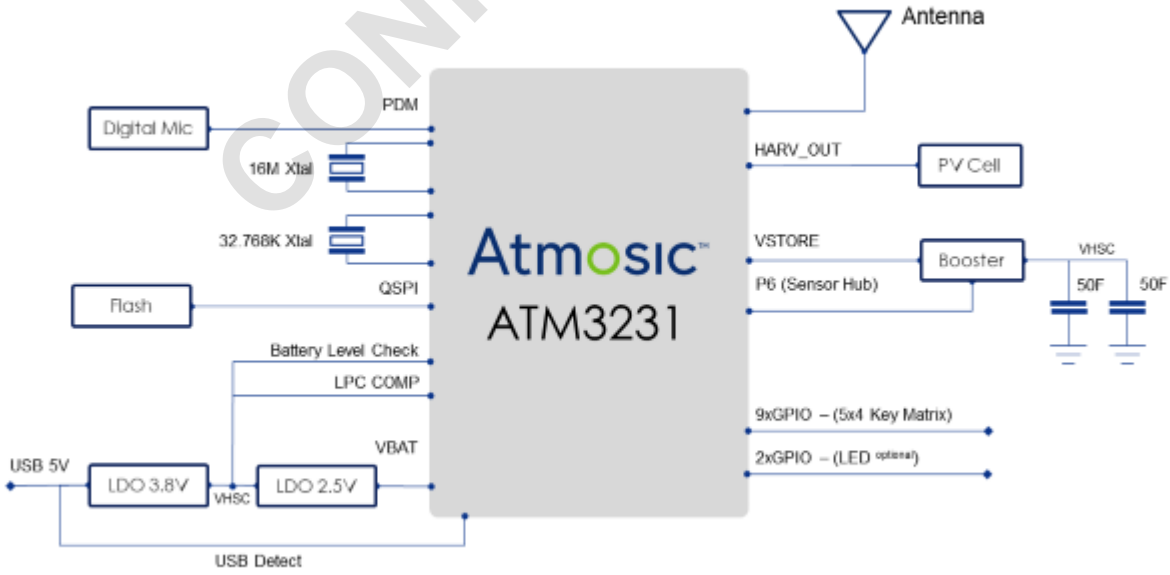


Figure 4 - Block Diagram for Hybrid Supercapacitor Configuration

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3. Android TV Device Support

The Atmosic PV Remote Control has been tested and verified with ADT-3 Android TV device and Onn Android TV device (from Walmart). The Android TV devices are not provided with the Atmosic PV Remote Control and are available separately. Refer to the References section for more information about the Android TV devices.

4. Remote Control Operation

4.1 General Operation

The state machine for the Remote Control is depicted in Figure 5. Upon power-up the Remote Control will automatically enter the pairing state if it has not already been paired. If previously paired it will attempt to reconnect and enter hibernation if it cannot connect in 1.28 seconds. From the pairing state it will timeout after 60 seconds if pairing is not completed. A keypress on the Remote Control is required to exit the hibernation state and restart the reconnection or pairing process.

Once connected the Remote Control has a default connection interval of 10 ms with a slave latency setting of 99.

The hybrid supercapacitor (ATMRC-3231-HSC) Remote Control has a voltage monitoring function that operates continuously except when the Remote Control is in the SoC Off state. If the voltage of the HSC is lower than VBAT_low voltage level (see Table 1), the system will stop working and enter Hibernation until the voltage rises above VBAT_low_hys from either PV cell harvesting or USB charging.

If the voltage falls below VBAT_min, the Remote Control will enter SoC Off and will not be operable again until it reaches VBAT_lpc via USB charging.

When the USB cable is plugged in for charging a hardware reset disables all Remote Control functionality until the USB cable is unplugged.

The critical low voltage levels are shown in Table 1.

VBAT	Voltage Level
VBAT_lpc	2.95 V
VBAT_low_hys	2.75 V
VBAT_low	2.65 V
VBAT_min	2.55 V

Table 1 - Critical Low VBAT Voltage Level

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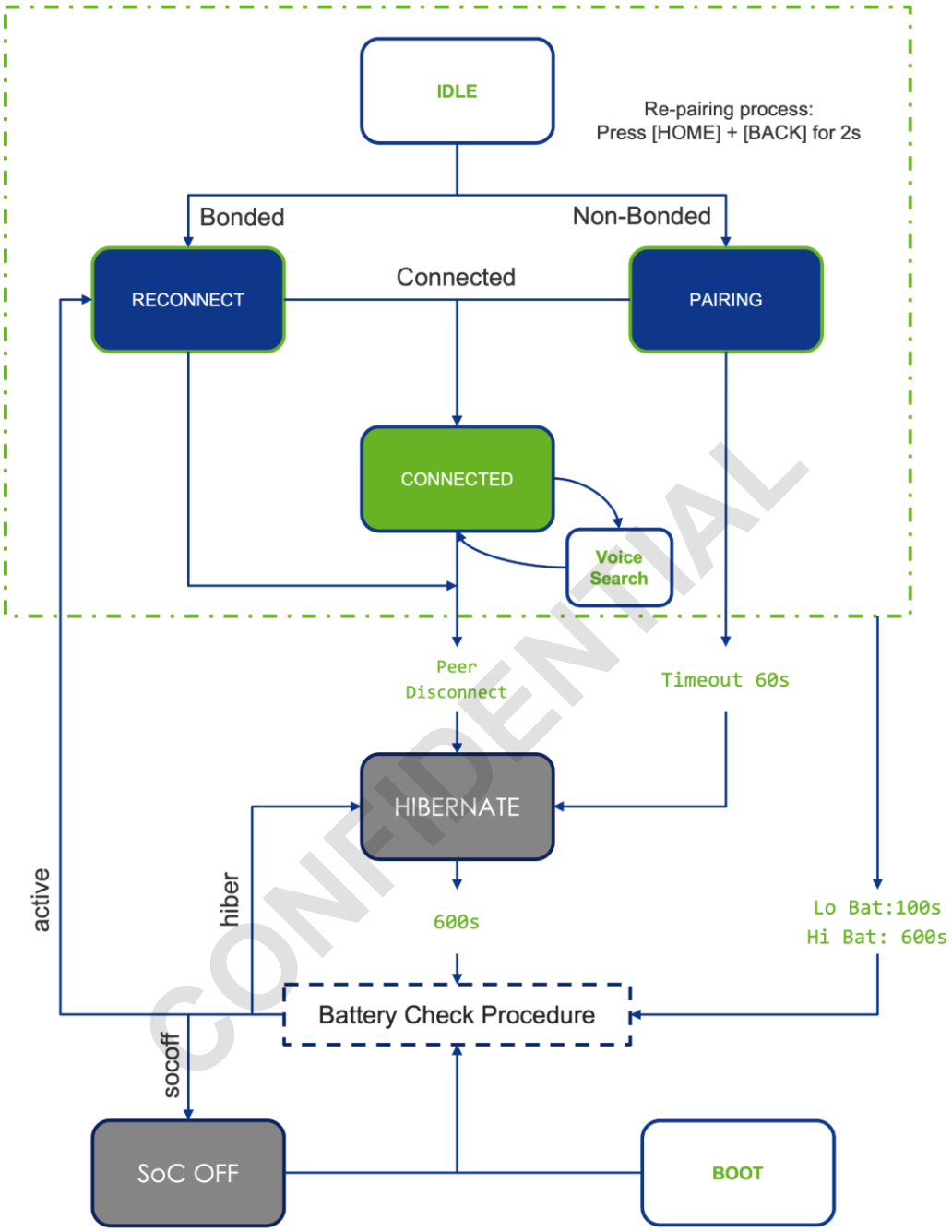


Figure 5 - Remote Control State Machine

4.2 Pairing the Remote Control

The Atmosic PV Remote Control is compatible with both the ADT-3 and onn. Android TV devices. To pair the remote, use the `connect a Bluetooth device` option in the `Settings` menu of the Android TV device. Use the Remote Control already paired to the Android TV device to select the `Atmosic RCU` device and begin the pairing process. If the `Atmosic RCU` device is not found it may either already be paired with another

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device such as a phone, or other TV device, or in the hibernate state. Press the `Home` and `Back` keys simultaneously for two seconds on the Atmosic PV Remote Control to enter pairing mode. The green LED blinks to indicate pairing is in progress.

If pairing is not completed in 60 seconds after exiting idle/hibernation the Atmosic PV Remote Control will enter the Hibernate state, press the `Home` and `Back` keys simultaneously for two seconds again to start the re-pairing process.

To pair with another TV device, delete the `Atmosic RCU` from the list of Bluetooth devices in the `Settings>connected devices>previously connected` menu of the Android TV device.

4.3 LED Indicator

The Remote Control supports LED indicator for the operating states listed in Table 2.

Operating State	LED Behavior
USB cable in and charging	Red LED on
USB cable in and charging full	Green LED on
Pairing	Green LED blinking at 250 ms interval
Reconnecting	Green LED blinking at 100 ms interval
Remote key press (connected)	Green LED on for 100 ms

Table 2 - LED Indicator Description

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4.4 USB Type-C Connector

The USB type-C connector is used to charge the hybrid supercapacitor as shown in Figure 6, it is not necessary to remove the back cover for charging.

When plugging in a USB cable, the Remote Control will reset and then enter hibernation. The solid red LED is displayed until the hybrid supercapacitor is fully charged and the green LED turns on.



Figure 6 - USB Type-C connection for Charging

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5. Remote Control Power Consumption

The Remote Control supports direct current consumption measurements to demonstrate the low operating current of the ATM3231 device for a Remote Control application.

The measured current consumption in three application states at 3 V is shown in Table 3. These measurements were taken at the battery connection of the CR2032 of the Remote Control during operation with the PV cell and supercapacitor disconnected.

Operating State	Current Consumption (3 V Supply)
Connection	4.89 µA
Single Key Press	105.42 µA
Voice Input	1.44 mA

Table 3 - Remote Control Power Consumption

Atmosic has developed various analysis tools to help customers estimate the daily energy consumption of the Remote Control based on a model of daily usage as well as tools to analyze the performance of the Remote Control based on different PV cell types and sizes; battery technology and size; supercapacitor size; and ambient light levels. Submit a support request for additional information about Atmosic analysis tools.

5.1 Measuring Power Consumption

5.1.1 CR2032 Coin Cell Version

It is possible to duplicate these measurements in the CR2032 coin cell version of the Remote Control by covering up the PV cell and disconnecting the supercapacitor as an energy storage device. The supercapacitor can be disconnected by removing the jumper on the bottom of the Remote Control as shown in Figure 7.

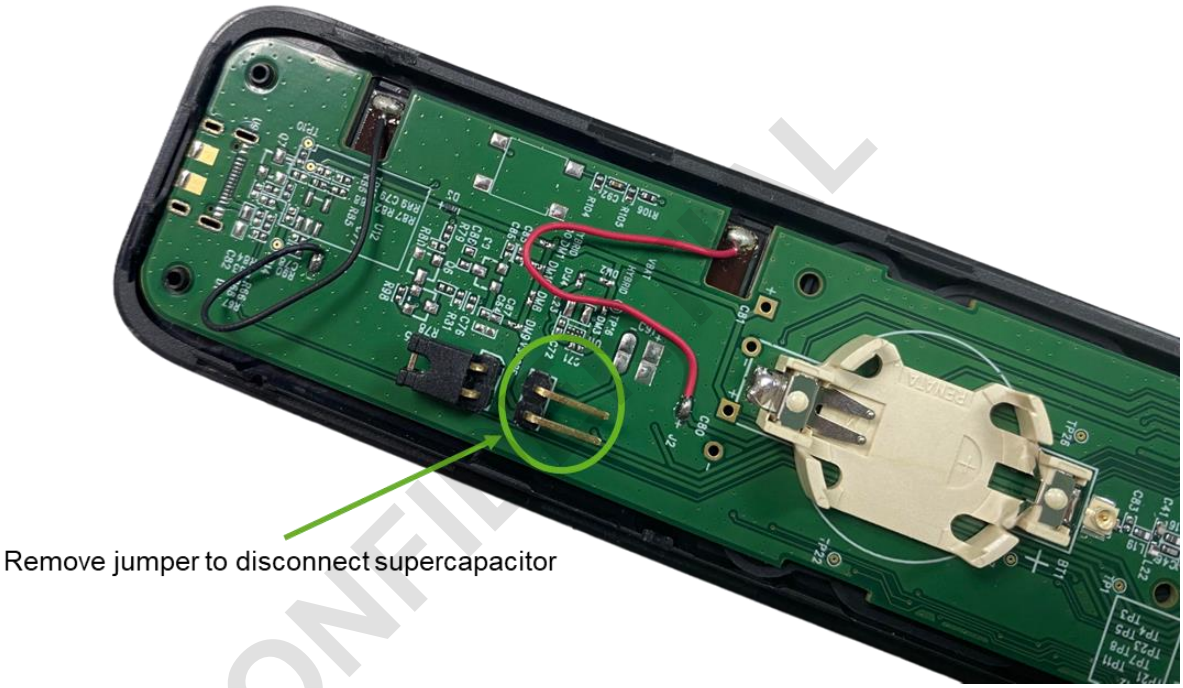


Figure 7 - Disconnect the Supercapacitor

Current consumption at the battery connection is measured across a separate jumper connection on the bottom side of the Remote Control. The jumper used to connect the CR2032 coin cell to the rest of the remote is shown in Figure 8. The current consumption of the entire Remote Control from the battery can be measured across this jumper.

Note: *If the supercapacitor jumper is installed and the supercapacitor contains sufficient stored energy for a Remote Control operation there will be no measurable current from the battery connection.*

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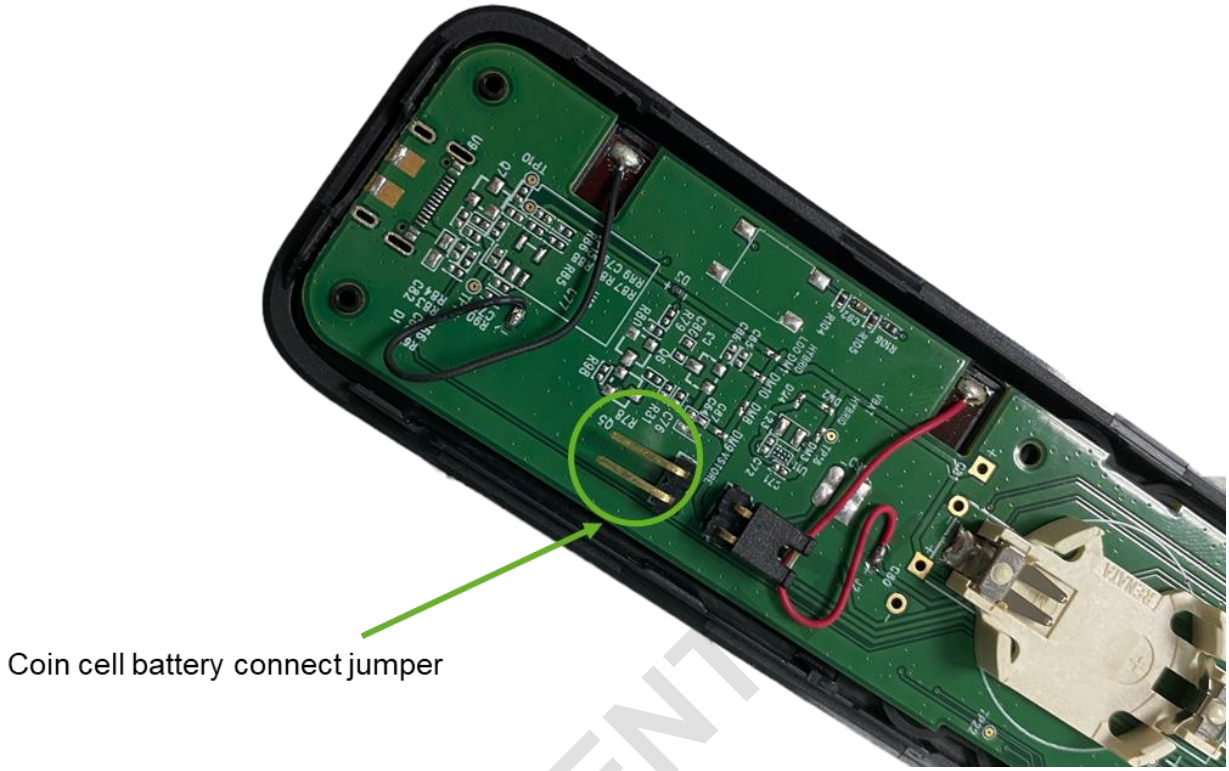


Figure 8 - Disconnect Coin Cell Battery from the Remote Control

5.1.2 Hybrid Supercapacitor Version

Making power consumption measurements with the hybrid supercapacitor version requires disconnecting the two 50 F hybrid supercapacitors on the bottom side and covering up the PV cell. These supercapacitors are connected to a switching regulator that powers the ATM3231 device.

To measure current consumption, de-solder the dummy pad DM3 to disconnect the hybrid supercapacitor from the ATM3231 device. The location of DM3 is shown in Figure 9. Power the Remote Control with a DC power analyzer to measure current consumption from switching regulator input. The current consumption of the Remote Control powered by hybrid supercapacitor at 3.8 V is shown in Table 4.

Operating State	Current Consumption (3.8 V Supply)
Connection	4.8µA
Single Key Press	89.16µA
Voice Input	1.31mA

Table 4 - Remote Control Current Consumption Powered by Hybrid Supercapacitor

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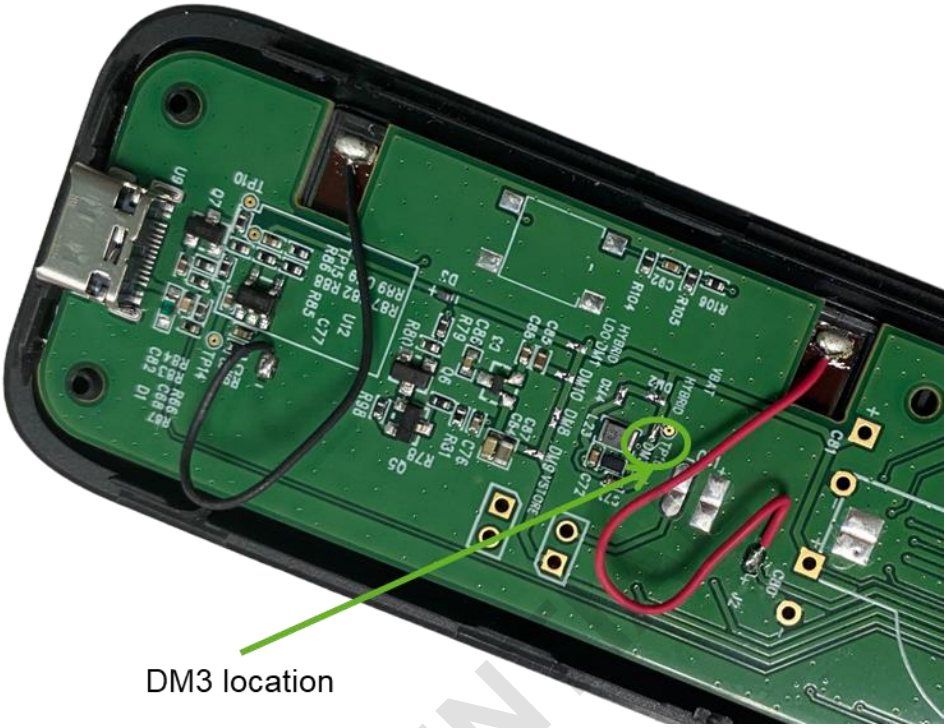


Figure 9 - DM3 Pad Location

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6. Upgrading the Remote Firmware

It is recommended that any firmware upgrades be done Over-the-Air with the Atmosic mobile application. Hardwired methods of upgrading the firmware are also available, but require removal of the unit's back housing. For additional information Contact Atmosic support.

6.1 Over-the-Air (OTA) Update via Mobile App

The Atmosic DevTools App supports OTA upgrading the remote firmware and is available on both Android and iOS. It can be downloaded via these links:

- Android
- iOS

Before upgrading, the Remote Control must be put into pairing mode. See Pairing the Remote Control section of this document for details.

For detailed instructions, please refer to the Firmware Upgrade Over-the-Air (OTA) section of the appropriate User Guide listed in the References section.

References

Title	Kit Part number	Description/Link
ATM3231 PV Remote Control	ATMRC-3231-HSC	Battery free configuration
ATM3231 PV Remote Control	ATMRC-3231-CC	Coin cell battery configuration
Kamcap 0.5 F Supercapacitor		Datasheet (3 V Winding Supercap)
VINATech 50 F Hybrid Supercapacitor		Datasheet - contact VINATech for additional information (matthewchoi@vinatech.com or tdickson@vinatechusa.com)
ADT-3 Android TV device		Android TV Guide (androidtv-guide.com) - ADT-3
onn. Android TV device		onn. Android TV 4K UHD Streaming Device with Voice Remote Control & HDMI Cable - Walmart.com
DevTools Mobile App User Guide, Doc. No. ATM-UGDTMA		Atmosic support_documentation page
ATM3231-PV_RC_V02_Battery Free_HDK, version V02 or later		Atmosic support website, submit a support request for access credentials

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Revision History

Date	Version	Description
November 28, 2022	0.50	Initial version created.

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